

SUMMERVILLE, SC, USA

WMO: 720632

Lat: 33.063N

Lon: 80.279W

Elev: 56

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00218

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.5	(c) 28.5	(d) 11.8	(e) 10.0	(f) 33.2	(g) 17.8	(h) 13.5	(i) 36.6	(j) 18.1	(k) 56.2	(l) 16.2	(m) 48.8	(n) 2.4	(o) 10	(p) 0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	95.3	77.9	92.7	77.0	90.7	76.4	80.8	89.2	79.6	87.3	78.6	86.0	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	150.7	83.5	77.4	142.8	81.9	77.0	141.0	81.7	44.1	89.7	43.1	87.3	42.2	85.7	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.2	(b) 12.4	(c) 10.8	(e) 20.2	(f) 98.2	(g) 5.3	(h) 2.2	(i) 16.4	(j) 99.8	(k) 13.2	(l) 101.0	(m) 10.2	(n) 102.2	(o) 6.4	(p) 103.8		
			DB													
			WB													
				18.7	82.4	4.6	1.3	15.4	83.3	12.7	84.1	10.1	84.8	6.7	85.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	47.0	52.7	56.2	64.7	72.3	78.9	80.8	80.2	76.1	66.0	55.0	51.7
	DBStd	13.77	9.53	9.34	8.56	6.86	5.46	4.05	2.97	2.98	4.30	7.64	8.00	9.49
	HDD50	411	170	73	37	1	0	0	0	0	0	1	34	95
	HDD65	2134	561	354	293	89	12	0	0	0	1	83	315	426
	CDD50	5957	78	149	230	441	692	866	954	936	784	496	184	147
	CDD65	2207	3	11	21	80	239	416	489	471	336	113	15	13
	CDH74	19540	14	92	217	762	2088	3928	4619	4224	2599	867	89	41
	CDH80	7942	0	5	34	170	738	1820	2128	1843	994	204	5	1
Wind	WSAvg	3.6	4.3	4.7	4.7	4.2	3.5	2.9	2.8	2.5	2.9	3.1	3.6	3.9
Precipitation	PrecAvg	47.0	3.3	3.6	3.7	3.0	3.4	5.2	5.3	5.9	4.1	3.5	2.6	3.4
	PrecMax	58.0	7.2	8.2	7.6	6.3	6.8	9.9	10.9	10.9	10.0	13.3	7.3	7.3
	PrecMin	34.4	1.3	1.4	0.5	0.9	1.1	2.5	2.6	3.2	0.3	0.0	0.2	0.8
	PrecStd	6.9	1.8	1.7	1.6	1.3	1.5	1.9	2.0	1.9	2.0	3.0	1.8	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	80.9	84.2	88.0	95.3	98.8	98.7	96.6	92.6	88.3	81.1	78.9
		MCWB	65.8	65.8	65.3	70.4	73.6	78.4	78.4	79.0	76.2	75.0	70.1	70.4
	2%	DB	70.7	76.9	80.8	84.3	90.3	95.1	95.3	93.1	90.4	85.6	77.1	73.3
		MCWB	63.0	64.6	65.1	68.7	71.6	77.2	78.7	77.8	75.7	74.1	67.2	67.3
	5%	DB	67.7	72.5	76.7	81.5	87.9	91.2	91.5	90.9	88.4	81.7	72.8	71.7
		MCWB	60.6	61.9	62.3	67.4	72.0	75.6	77.9	77.4	75.0	70.9	64.9	67.1
	10%	DB	63.4	68.5	72.2	79.0	84.4	90.1	90.4	89.9	86.2	79.2	70.2	68.1
		MCWB	58.7	60.7	61.2	65.7	71.5	75.4	77.6	77.1	74.2	69.6	64.3	64.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	69.2	69.8	74.0	76.1	81.4	82.7	81.4	79.5	78.0	73.1	72.9
		MCDB	71.4	74.6	75.8	80.9	85.0	90.4	93.8	89.2	84.9	82.1	77.3	76.0
	2%	WB	66.0	67.0	67.7	72.4	74.8	79.5	81.0	80.5	78.3	76.2	70.3	70.5
		MCDB	70.8	73.9	75.1	78.9	83.6	89.2	90.1	87.8	84.1	82.0	72.9	72.4
	5%	WB	63.3	64.5	65.8	70.5	73.7	78.1	79.7	79.4	77.3	74.3	68.3	68.1
		MCDB	64.6	70.3	73.7	76.7	82.1	87.5	88.0	86.4	83.6	79.6	71.4	69.9
	10%	WB	59.4	62.3	63.5	69.0	72.9	77.0	78.5	78.5	76.3	72.7	65.3	64.0
		MCDB	62.3	66.4	68.3	75.0	80.9	86.2	86.4	85.2	83.0	77.1	70.0	66.2
Mean Daily Temperature Range	5% DB	MDBR	21.7	22.3	23.8	24.2	21.6	20.6	19.3	18.3	18.9	21.5	23.0	20.2
		MCDBR	25.7	25.9	27.6	26.9	26.6	23.5	21.7	20.7	21.7	22.5	24.9	22.3
		MCWBR	16.8	14.5	13.1	12.2	10.1	7.8	8.0	7.6	8.3	11.3	15.6	14.7
	5% WB	MCDBR	19.8	22.6	21.6	20.7	20.4	20.7	20.0	18.7	17.7	18.2	18.5	18.3
		MCWBR	16.1	14.7	11.5	10.8	8.6	7.8	8.1	7.6	7.6	9.7	13.7	13.5
Clear-Sky Solar Irradiance	taub		0.324	0.333	0.366	0.408	0.471	0.507	0.544	0.520	0.437	0.392	0.346	0.328
	taud		2.527	2.497	2.423	2.293	2.160	2.111	2.023	2.085	2.312	2.411	2.500	2.548
	Ebn at Noon		284	292	289	280	262	252	241	245	263	267	273	276
	Edh at Noon		27	30	35	42	48	50	55	51	39	33	27	25
All-Sky Solar Radiation	RadAvg		906	1112	1463	1859	1999	2039	1963	1772	1514	1300	1004	804
	RadStd		49	113	125	123	141	142	123	105	128	145	73	67

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+371	+259

Nomenclature: See separate page